Lab 20

Creating a Custom Policy

WSO2 API Manager 4.7.0 · Practical exercise

Before You Begin

URLs in this guide use placeholders. Replace them with the values of the API Manager you are using:

- **`<APIM_HOST>`** – host name or IP address of the API Manager server (localhost when API Manager runs on your own laptop).
- **`<HTTPS_PORT>`** – API Manager HTTPS port: 9443 plus any port offset (Publisher, Developer Portal).
- **`<GW_HTTPS_PORT>`** – API Gateway HTTPS port: 8243 plus any port offset.

The server uses a self-signed certificate by default, so the browser shows a security warning the first time: choose Advanced and continue. The -k option in the curl command skips the same check.

Training Objective

Learn how to create a custom common policy (a reusable piece of Gateway mediation logic) and attach it to an API's request or response flow in the Publisher.

Business Scenario

PizzaShack wants every response from its API to carry a header that shows it went through the API Gateway, without changing the back end.

High-Level Steps

- Create a common policy that adds a header
- Attach it to the PizzaShack API's response flow and deploy
- Call the API and check the new response header

Prerequisites

- The PizzaShack API is deployed and published (context `/pizzashack`, version `1.0.0`).
- An application subscribed to the PizzaShack API, with production keys generated (from the earlier labs).
- If you did Lab 19, its clean-up is done: the original `velocity_template.xml` is restored and the custom handler jar is removed.
- Lab file: `sampleAddHeader.j2`.

Part 1 – Create the Policy File

1. Create a file named `sampleAddHeader.j2` with this single line (also provided in the lab files):

sampleAddHeader.j2

```
<property action="set" name="{{headerName}}" value="{{headerValue}}"
scope="transport" />
```

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The `{{ }}` parts become policy attributes. You fill them in when you attach the policy to an API, so the same policy can add any header. `scope="transport"` means the value is sent as an HTTP header.

Part 2 – Add the Policy in the Publisher

2. Log in to the Publisher, click **Policies** in the left menu and click **Add New Policy**.

Publisher URL

```
https://<APIM_HOST>:<HTTPS_PORT>/publisher
```

3. Fill in **General Details**:

Field	Value
Name	Sample Add Header
Version	1
Description	Adds a header
Applicable Flows	Request and Response

4. In **Gateway Specific Details**:

- Select the regular (Synapse) gateway.
- Click **Upload Policy File** and upload `sampleAddHeader.j2`.
- Keep **Supported API Types** as HTTP.

5. In **Policy Attributes**, click **Add Policy Attribute** twice and enter the two attributes below. Copy the regular expressions exactly.

Name	Display Name	Required	Type	Validation Regex
`headerName`	Header Name	On	String	See regex 1 below
`headerValue`	Header Value	On	String	See regex 2 below

Regex 1 – headerName:

```
^([a-zA-Z_][a-zA-Z\d_\-\ ]*)$
```

Regex 2 – headerValue:

```
^([a-zA-Z\d_][a-zA-Z\d_\-\ ]*)$
```

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6. Click **Save**. The policy appears in the common policy list and any API can use it.

Part 3 – Attach the Policy to the PizzaShack API

7. Open the PizzaShack API and click **Policies** in the left menu. Select **API Level Policies**. In the policy list on the right, find **Sample Add Header** and drag it into the **Response Flow**.

8. In the form that opens, enter the values below and click Save:

Field	Value
Header Name	`X-PizzaShack-Served-By`
Header Value	`WSO2-APIM`

9. Click **Save and deploy** at the bottom of the Policies page and deploy the new revision to the Gateway.

Part 4 – Test the Policy

10. Get an access token: in the Developer Portal, open your application, go to **Production Keys**, click **Generate Access Token** and copy the token.

    Developer Portal URL

    ```
    https://<APIM_HOST>:<HTTPS_PORT>/devportal
    ```

11. Call the API and show the response headers (`-i`). Replace `<ACCESS_TOKEN>` with the token you copied:

    macOS / Linux

    ```
    curl -k -i \
      -H "Authorization: Bearer <ACCESS_TOKEN>" \
      https://<APIM_HOST>:<GW_HTTPS_PORT>/pizzashack/1.0.0/menu
    ```

    Windows (Command Prompt)

    ```
    curl -k -i ^
      -H "Authorization: Bearer <ACCESS_TOKEN>" ^
      https://<APIM_HOST>:<GW_HTTPS_PORT>/pizzashack/1.0.0/menu
    ```

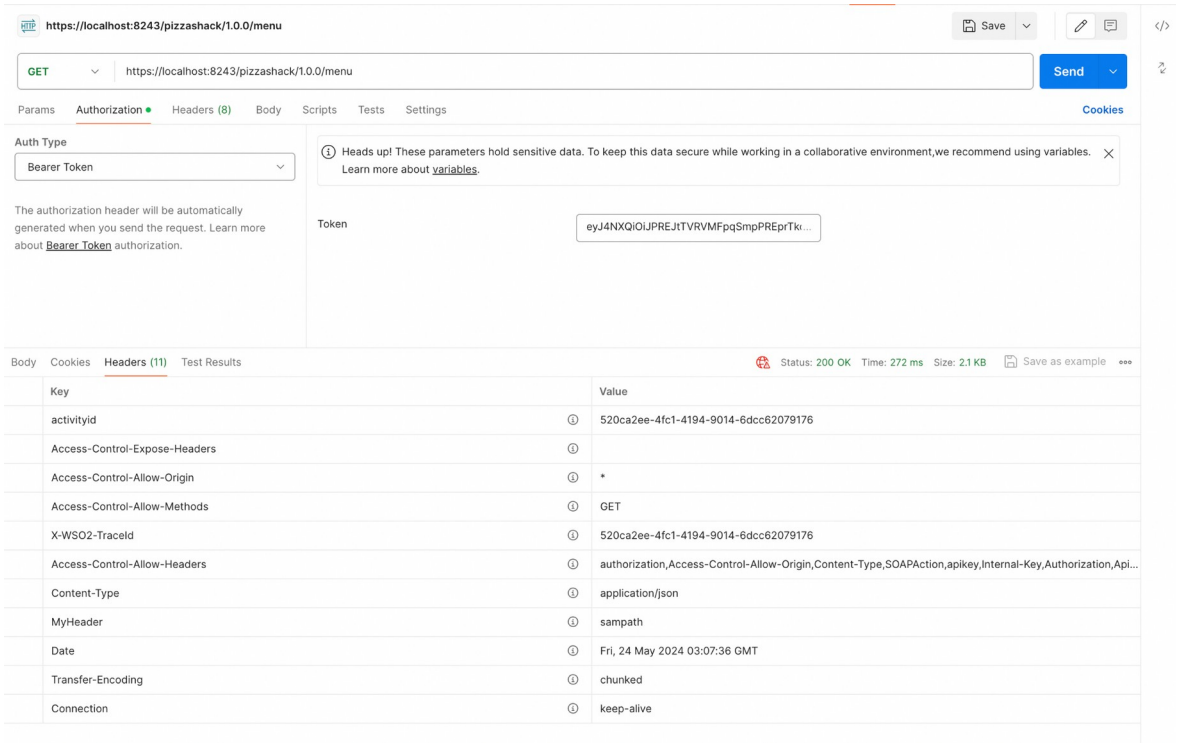
    Expected: `HTTP/1.1 200 OK`, and among the response headers:

    ```
    X-PizzaShack-Served-By: WSO2-APIM
    ```

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12. Using Postman instead: send `GET https://<APIM_HOST>:<GW_HTTPS_PORT>/pizzashack/1.0.0/menu` with Authorization type **Bearer Token** and the token, then open the **Headers** tab of the response.

    Example of a custom response header in Postman (the header name and value in the screenshot differ: `MyHeader: sampath`)



Expected Outcome

A reusable common policy has been created and attached to the PizzaShack API's response flow, so every response carries the X-PizzaShack-Served-By: WSO2-APIM header. The same policy can be attached to other APIs with different header names and values.

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